(Please write your Enrollment Number) **Enrollment No._____**

End-Term Examination
(CBCS)(SUBJECTIVE TYPE)(OffLine)
Course Name: Business Mathematics, Semester: 1
(December, 2024)

Subject Code: AMC-101	**Subject: Business Mathematics**
Time : 3 Hours	**Maximum Marks :60**

Note:Q1 is compulsory. Attempt one question each from the Units I, II, III & IV.

Q1	**Choose the correct alternative:**	**(2.5*8 =20)**	**CO Mapping**
	a) The number of 3-digit even numbers that can be formed with no digit repeated by using the digits 0,1,2,3,4, and 5 is: (i) 48 (ii) 50 (iii) 52 (iv) 54		**CO1**
	b) The fourth term of an arithmetic series is 14 and eighth term is 26. Then the sum of first 10 terms is: (i) 155 (ii) 165 (iii) 175 (iv) 185		**CO1**
	c) The determinant of the matrix $\begin{bmatrix} -1 & 5 & -2 \\ 2 & 1 & 0 \\ 2 & 7 & 3 \end{bmatrix}$ is: (i) 57 (ii) -57 (iii) 60 (iii) -60		**CO2**
	d)The demand equation for the commodity A is $D_A = 85 - 3\,P_A + 9P_B$ and the respective supply production is $S_A = -5 + 15\,P_A$ The market equilibrium price for the commodity A is (i) 6 (ii) 10 (iii) 5 (iv) 6		**CO2**
	e)The total revenue in Rupees received from the sale of x units of a product is given by $R(x) = 3x^2 + 36x + 5$. Then the marginal revenue when $x = 5$ is: (i) 60 (ii) 66 (iii) 68 (iv) 62		**CO3**
	f)If $q = 2p^2$, where q is the quantity demanded and p is price, then the price elasticity of demand is (i) 2 (ii) 4 (iii) 8 (iv) 10		**CO3**

	g)The marginal revenue function of a firm is given by $MR = 240 - 4x$. Then the total revenue function is: (i) $240x - 2x^2$ (ii) $240x - 2x^2 + c$ (iii) $340x - 3x^2$ (iv) $340x - 3x^2 + c$		**CO4**
	h)The solution of the integral $\int_0^3 (3x^2 + 5x + 2)dx$ is (i) 50 (ii) 55.5 (iii) 60 (iv) 62.5		**CO4**
	UNIT I		**CO Mapping**
Q2	Prove the following results using the Principle of Mathematical Induction: (a) $n + n^2 is\ an\ even\ natural\ number, n \epsilon N$ (b) $3^n - 1$ is divisible by 2 for each $n \epsilon N$	**(10)**	**CO1**
Q3	A sum of Rs 6,240 is paid off in 30 installments, such that each installment is Rs 10 more than the preceding installment. Calculate the value of the first installment.	**(10)**	**CO1**
	UNIT II		**CO Mapping**
Q4	If A = $\begin{bmatrix} 3 & 2 \\ 4 & 1 \end{bmatrix}$ and B = $\begin{bmatrix} a & b \\ 3 & 5 \end{bmatrix}$, find a and b such that AB = BA	**(10)**	**CO2**
Q5	Let A = $\begin{bmatrix} 3 & 4 \\ 3 & 2 \end{bmatrix}$, B = $\begin{bmatrix} 1 & 3 \\ -2 & 5 \end{bmatrix}$ and C = $\begin{bmatrix} -2 & 5 \\ 3 & 4 \end{bmatrix}$, find the following (a) 3A –C (b) AB	**(10)**	**CO2**
	UNIT III		**CO Mapping**
Q6	Find the price elasticity of demand for the following function $q = 200 - 4p$, at q = 40	**(10)**	**CO3**
Q7	The short-run production function of a manufacturer is given as $q = 11L + 16L^2 - L^3$. Find the average product function and marginal product function, and find L at which the two are equal (Note: q is Total Product)	**(10)**	**CO3**
	UNIT IV		**CO Mapping**
Q8	Evaluate the following integrals: (a) $\int 3x^2 + 5x - 2\, dx$ (b) $\int e^{-2x}\, dx$	**(10)**	**CO4**
Q9	A firm's marginal revenue function is MR(q)=100−5q. Find the total revenue function. Find the quantity at which total revenue will be 0.	**(10)**	**CO4**